

Healthcare Predictive Analytics

30-Day Hospital Readmission Risk Classification

Submission-ready analysis using the supplied healthcare patient sample. The model uses a Random Forest classifier with class-weight balancing and 3-fold stratified out-of-fold evaluation. Blood pressure was converted from text into systolic and diastolic numeric features.

Dataset rows	10
Positive readmissions	3
Negative cases	7
Model	Random Forest (300 trees, class_weight='balanced')
Evaluation	3-fold stratified out-of-fold predictions
Threshold	0.50 probability

1. Objective

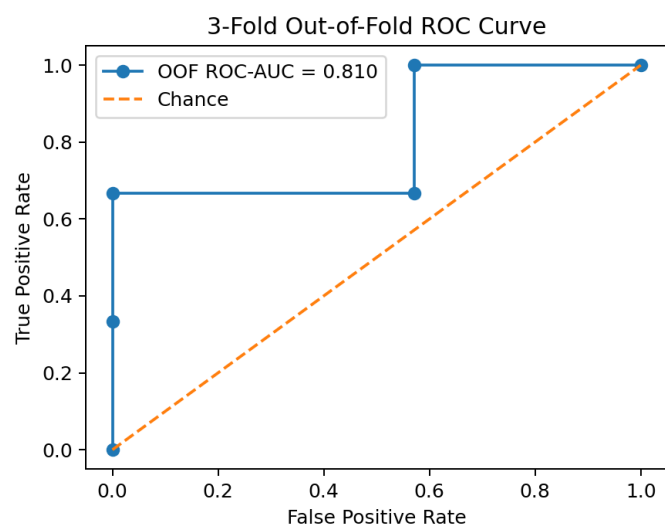
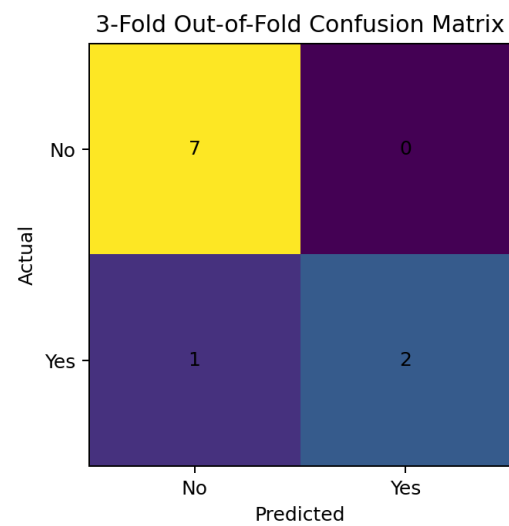
Build and evaluate a machine-learning classification model that estimates whether a patient will be readmitted within 30 days, based on the attributes available in the supplied dataset.

2. Methodology

- Removed PatientID from model features because it is an identifier rather than a clinical predictor.
- Converted BloodPressure values such as 145/92 into separate Systolic and Diastolic variables.
- One-hot encoded categorical variables: Gender, Diagnosis and PrimaryMedication.
- Used Random Forest with class_weight='balanced' to reduce the effect of the 7:3 class imbalance.
- Used 3-fold stratified out-of-fold evaluation because the dataset contains only three positive cases.
- Generated final risk probabilities by refitting the model on all supplied observations.

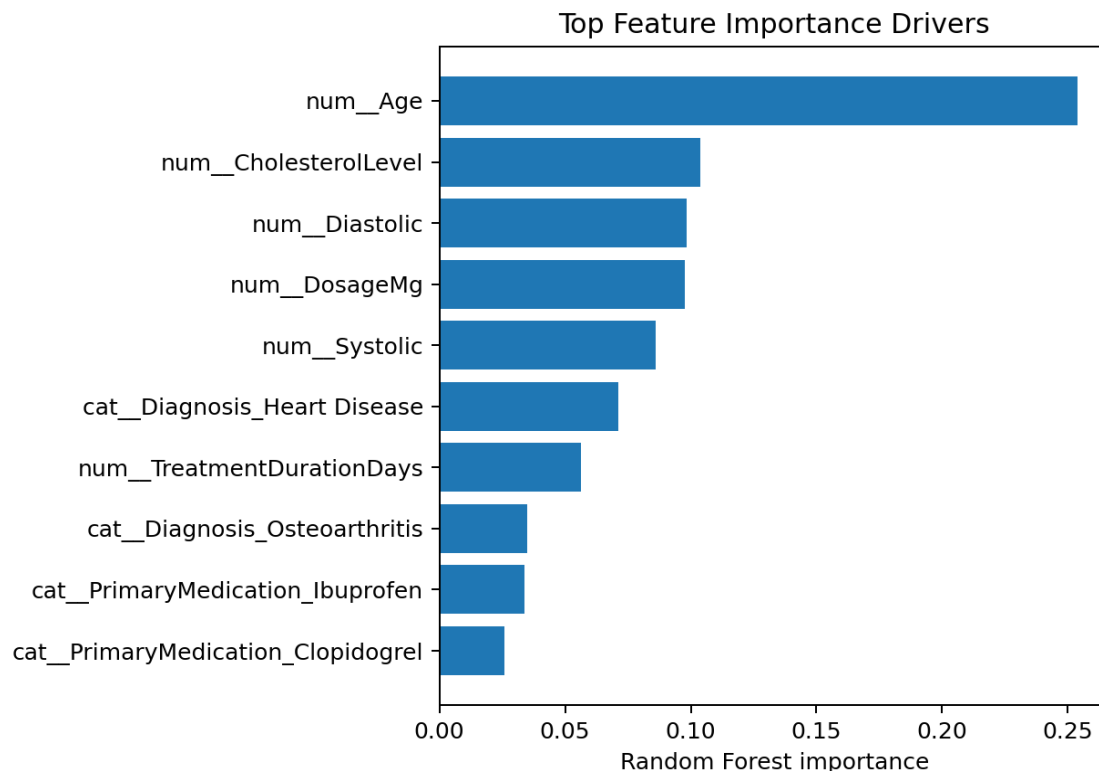
3. Model Evaluation

Metric	Value
Accuracy	0.900
Precision	1.000
Recall (Sensitivity)	0.667
F1-score	0.800
ROC-AUC	0.810



4. Feature Importance

The chart below shows the Random Forest's impurity-based feature importance after preprocessing. Because the dataset is very small, these importance values should be treated as exploratory rather than as evidence of clinical causation.



5. Prediction Output

The accompanying readmission_predictions.csv contains a probability, binary prediction and risk category for every supplied patient. Risk categories are defined as Low (<0.33), Moderate ($0.33\text{--}0.66$) and High (≥ 0.66). These categories are project-level thresholds, not validated clinical decision thresholds.

6. Key Findings

The supplied sample contains 10 patients, including 3 readmitted and 7 not readmitted cases. The leading model-derived feature drivers were: num__Age, num__CholesterolLevel, num__Diastolic, num__DosageMg, num__Systolic. The exact ordering can change substantially with a larger dataset or a different random split.

7. Limitations

- Only 10 patient records are available, with only 3 positive readmission cases.
- The dataset does not provide a separate validated 30-day follow-up field; the Readmitted field is treated as the target requested by the task.
- No external validation cohort is available.
- Model metrics are unstable with such a small sample and must not be interpreted as clinical performance.
- The predictions are for educational/project use and are not clinical recommendations.

8. Conclusion

A complete reproducible machine-learning workflow was implemented for readmission-risk classification. The workflow includes preprocessing, imbalance-aware training, out-of-fold evaluation, feature importance analysis and patient-level prediction export. For meaningful clinical research, the next step would be evaluation on a much larger, independently collected dataset with a clearly defined 30-day readmission outcome.

References / Technical Resources

- Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45, 5–32.
- Kuhn, M. & Johnson, K. (2013). *Applied Predictive Modeling*. Springer.